

$$F = [1 \ 0 \ 0 \ 0 \ 0 \ 1]^T$$

$$\therefore \|f\| = 1$$

$$\therefore h = [1 \ 0 \ 0] \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = f^T \chi$$

$$= 0$$

$$q = \hat{f} \chi = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}$$